Geophys. J. Int. (2009) **178,** 1327–1336 doi: 10.1111/j.1365-246X.2009.04222.x

New palaeomagnetic data supporting the extent of the stable body of the South China Block since the Cretaceous and some implications on magnetization acquisition of red beds

Yosuke Tsuneki,[1] Hayao Morinaga[1] and Yuyan Liu[2]

[1]*Department of Global Tectonics, Graduate School of Life Science, University of Hyogo, Shosha* 2167, *Himeji* 671–2280, *Japan. E-mail: magnet@sci.u-hyogo.ac.jp*

[2]*Faculty of Earth Sciences, China University of Geosciences (Wuhan), Yujiashan, Wuhan* 430074, *People's Republic of China*

Accepted 2009 April 22. Received 2009 April 8; in original form 2008 July 10

SUMMARY

Previous palaeomagnetic studies have demonstrated that a large part of the South China Block (SCB) has behaved as a stable body since the Cretaceous. We undertook a palaeomagnetic investigation of Lower Cretaceous red sandstones at 24 sites within the Ganzhou and Xingguo basins in southern Jiangxi Province in China. The aim of this study was to further constrain the extent of the SCB, which has been stable since the Cretaceous. We isolated the characteristic directions of higher temperature components (HTCs) with an unblocking temperature from 650 to 700 °C, by progressive thermal demagnetization and principal component analysis. The optimal concentration of site mean HTC directions calculated using the direction-correction tilt test was achieved at 51.2 ± 32.4 per cent untilting, indicating syntilting magnetization, suggesting that the remanences were not acquired immediately after sedimentation. Most Cretaceous sedimentary basins on the eastern part of the SCB were controlled by fault movement (extensional basins). Most tilting in such an extensional basin is thought to have progressed contemporaneously with the structural and stratigraphic development of the basin. We concluded that the remanence acquisition of red sandstones on the Ganzhou and Xingguo basins occurred during synsedimentary tilting. We adopted 51.2 per cent untilted directions of the HTCs as the palaeomagnetic field directions during the Early Cretaceous. The mean palaeomagnetic pole (76.3°N, 224.3°E, $\alpha_{95} = 3.3°$), calculated using virtual geomagnetic poles from 23 sites (excluding site 20, which has an insufficient number of samples showing a stable magnetic component), is in good agreement with the Late Cretaceous palaeomagnetic poles from the same region and also with most of Cretaceous poles previously reported from the stable body of the SCB. This agreement indicates that the Jiangxi region has been part of the stable body of the SCB since the Cretaceous. This result also demonstrates that the tectonic influence of the India–Asia collision did not destroy the rigidity of most of the SCB. The global mean pole position (78.8°N, 214.4°E, $\alpha_{95} = 2.6°$), calculated using sixteen Cretaceous poles from the stable body of the SCB, is suitable for use as the reference Cretaceous palaeomagnetic pole of the stable SCB.

Key words: Magnetic mineralogy and petrology; Palaeomagnetism applied to tectonics; Palaeomagnetism applied to geologic processes; Continental tectonics: extensional; Dynamics and mechanics of faulting; Asia.

1 INTRODUCTION

The India–Asia collision caused tectonic deformation of eastern and southeastern Asia, especially of the Indochina Block. Previous palaeomagnetic investigations (Huang & Opdyke 1992, 1993; Funahara *et al*. 1993; Chen *et al*. 1995; Sato *et al*. 1999, 2007, data o–u in Fig. 1) reported that some areas of Qiangtang Terrane and Indochina, excluding the Song Da Terrane (Takemoto *et al*. 2005, datum v in Fig. 1), have been extruded to the southeast and/or rotated clockwise. Palaeomagnetic investigations (Funahara *et al*. 1992; Gilder *et al*. 1993; Otofuji *et al*. 1998; Liu & Morinaga 1999, data g, h, m and n in Fig. 1) also revealed that the India–Asia collision caused small and differential tectonic deformation of the regions along the Red River Fault (RRF) within the South China Block (SCB), as shown by the dashed ellipse in Fig. 1. Liu & Morinaga (1999) demonstrated that the 400-km-wide swath along the RRF within the SCB was extruded to the southeast due to the extrusion of the Indochina Block. The SCB, therefore, is an ideal

© 2009 The Authors
Journal compilation © 2009 RAS

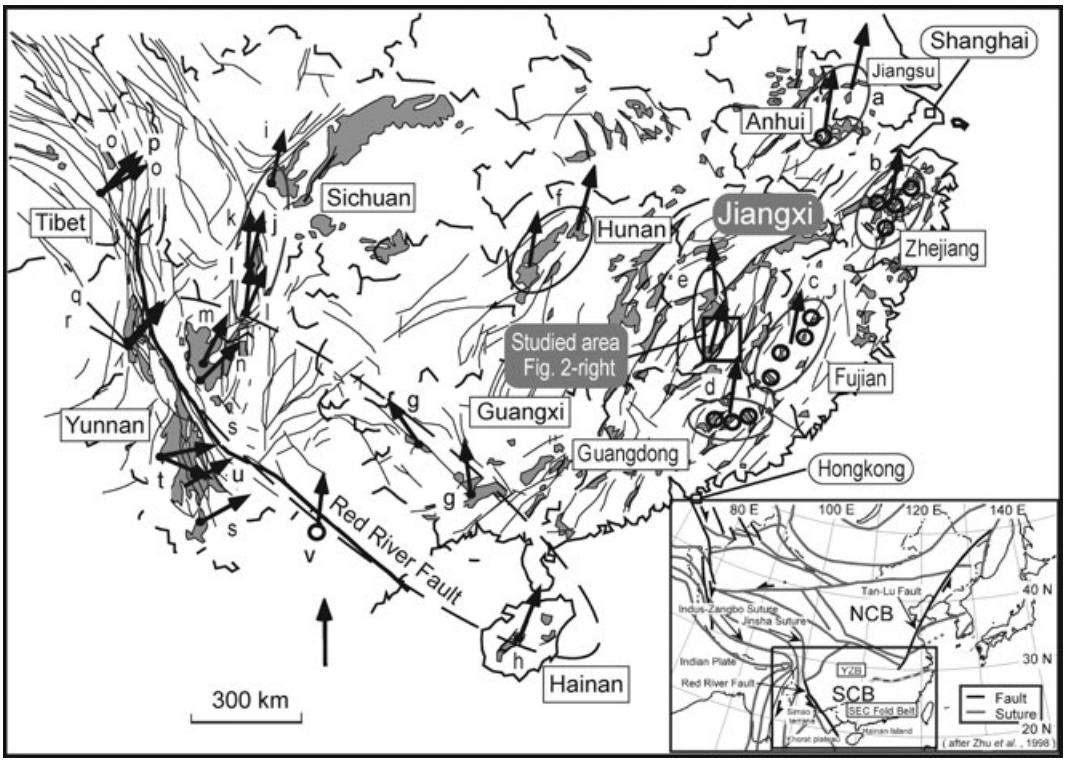


Figure 1. A simplified geological map of the South China Block (SCB) and adjacent areas, showing the distribution of faults and Cretaceous basins (MGMRC 1990). The inset tectonic map of East Asia from Zhu *et al*. (1998) shows the location of the main map. Arrows show the mean declinations reported in previous studies described in the text. The large ellipse in the southwestern SCB shows slightly deformed regions proposed by previous studies (Funahara *et al*. 1992; Gilder *et al*. 1993; Otofuji *et al*. 1998; Liu & Morinaga 1999).

region for investigating the tectonic evolution of continental deformations caused by the India–Asia collision (Tapponnier *et al*. 1982; Peltzer & Tapponnier 1988; Replumaz & Tapponnier 2003) and/or a sinistral ductile shear that occurred during the Mesozoic and Cenozoic (Xu *et al*. 1987; Li *et al*. 2001).

Meanwhile, the results of palaeomagnetic studies (Lin 1984; Kent *et al*. 1986; Zhu *et al*. 1988; Otofuji *et al*. 1990; Enkin *et al*. 1991; Huang & Opdyke 1992; Gilder *et al*. 1999; Morinaga & Liu 2004; Tamai *et al*. 2004; Sun *et al*. 2006; Zhu *et al*. 2006; Wang & Yang 2007, data a–f and i–l in Fig. 1) indicate that most of the SCB suffered no apparent effects from the India–Asia collision (Fig. 1). Morinaga & Liu (2004) reported palaeomagnetism of Cretaceous red sandstones was distributed over the Zhejiang, Fujian and Guangdong Provinces of the eastern SCB. They found good agreement among the palaeomagnetic poles of these coastal regions and also between these poles and the Cretaceous mean pole from the Sichuan Province of the western SCB, as determined by Gilder *et al*. (1993). They concluded that there is no relative tectonic movement between the eastern and western parts of the SCB, excluding the 400-km-wide swath along the RRF (Liu & Morinaga 1999), and the parts have behaved as a stable body since the Cretaceous. They also found a good agreement among the mean Cretaceous poles of the stable body of the SCB, the North China Block (NCB) and stable Eurasia. Furthermore, Zhu *et al*. (2006) reported that palaeomagnetism of Cretaceous red sandstones was distributed in western Hunan Province in the central SCB, where palaeomagnetic data had been scarce. They found good agreement between the Cretaceous palaeomagnetic pole of the Hunan region and previously reported Cretaceous poles from the eastern and western SCB. They concluded that the Hunan region, and thus the central SCB, also was part of the stable body of the SCB since the Cretaceous, supporting the idea of Morinaga & Liu (2004).

On the other hand, Wang & Yang (2007) reported palaeomagnetism of Upper Cretaceous red sandstones distributed in southern Jiangxi Province, in the eastern part of the SCB. They demonstrated that Late Cretaceous poles from the SCB do not cluster. They also found that the distribution of these poles can be separated two groups and is generally related to their sampling locations, situated in coastal areas and western inland areas of the SCB. The pole-distribution separation means that there is a significant difference in palaeolatitude ($9.9^\circ \pm 4.1^\circ$) between coastal and inland areas. They presented two possible interpretations for this separation. The first interpretation is that eastern coastal areas of the SCB were displaced away from the inland areas due to the NE-oriented shearing fault system. However, as these faults are inferred to be sinistral strike-slip faults that have been active since the Late Cretaceous (Xu *et al*. 1987; Li *et al*. 2001), the southerly displacement of the coastal areas is countered by geological observations. The second interpretation is that the separation may be related to the problem of inclination shallowing in red beds. Consequently, they demonstrated that it is difficult to declare the whole SCB is a tectonically stable block.

For a better understanding of continental deformations' tectonic evolution, it is necessary to obtain more palaeomagnetic data and then to establish the extent of the stable body of the SCB. In this paper, we present the new palaeomagnetic results of the Lower Cretaceous red sandstones distributed in southern Jiangxi Province and discuss palaeomagnetic constraints on the extent of the stable body of the SCB. We also present some implications on magnetization acquisition of red beds.

2 GEOLOGICAL SETTING, SAMPLING AND LABORATORY PROCEDURES

Cretaceous red sandstones, thought to be good palaeomagnetic research materials because of their stable remanence, are widely distributed within the SCB. Most of these red sandstones were deposited in extensional basins during the Mesozoic (Xu *et al*. 1987). Most of the tilting in the extensional basins is thought to have occurred contemporaneously with the structural and stratigraphic growth of the basins (Schlische 1992). Such timing explains the large thickness (>1000 m) of sedimentary layers deposited in basins within the eastern SCB. These basins are believed to date to the Cretaceous on the basis of fossil vertebrates, pelecypods, ostracods, gastropods, plants and dinosaur eggs (Gao *et al*. 1999).

Sedimentary basins that contain red beds are widely distributed throughout Jiangxi Province (Fig. 1); the Ganzhou and Xingguo basins are located in the southern part of Jiangxi Province. We collected palaeomagnetic samples of red sandstone from 24 sites in the Ganzhou and Xingguo basins (Fig. 2), and sampled from the Maodian Formation at 11 sites in Xingguo County, three sites in the city area of Ganzhou and 10 sites along the main road running through the middle of the basin from Xingguo to Ganzhou Cities. The age of the Maodian formation is estimated to be Early Cretaceous (Gao *et al*. 1999). Our sampling sites are located to the east or northeast of the sites where Wang & Yang (2007) collected palaeomagnetic specimens from the Ganzhou formation (K2). Although the Maodian formation where we sampled is classified in the early stage of the Late Cretaceous (K_2^1) in fig. 1 of Wang & Yang (2007), we adopt the Early Cretaceous as the age of Maodian formation (Gao *et al*. 1999).

Orientation of palaeomagnetic block samples was determined using a magnetic compass. The latitude and longitude of the sites were determined using GPS. The present geomagnetic direction at each site was calculated using the International Geomagnetic Reference Field (Mandea & Macmillan 2000). In the laboratory, core specimens of 25 mm in diameter and 25 mm in length were drilled from each block sample. All the remanent magnetizations of samples were measured using a Natsuhara Giken (Osaka, Japan) SMM-85 spinner magnetometer. To evaluate the magnetic stability of samples and eliminate viscous magnetic components, progressive thermal demagnetization was conducted using a Natsuhara Giken TDE-91C thermal demagnetizer. The direction of characteristic remanent magnetization (ChRM) was computed using principal component analysis (Kirschvink 1980).

To identify magnetic carriers, isothermal remanent magnetization (IRM) was progressively acquired up to a maximum DC field of 2.8 T using a Magnetic Measurements (Germany) MMPM-10 pulse magnetizer. We also carried out thermal demagnetization of the composite IRMs with three different fields applied along three orthogonal axes (2.8, 0.4 and 0.12T in *Z*, *Y* and *X* axes, respectively; Lowrie 1990) and stepwise alternating field (AF) demagnetization for several pilot specimens. The composite IRMs were thermally demagnetized up to 680–700 °C, using the same instrument as for the progressive thermal demagnetization. Stepwise AF demagnetization was performed up to 180 mT using a Natsuhara Giken DEM-95C AF demagnetizer.

3 RESULTS OF MAGNETIC MEASUREMENTS

3.1 Thermal demagnetization and direction-correction (DC) tilt test

Typical examples of thermal and AF demagnetizations are shown in Fig. 3's orthogonal plots (Zijderveld 1967). Almost all red sandstone specimens generally showed stable behaviour throughout the

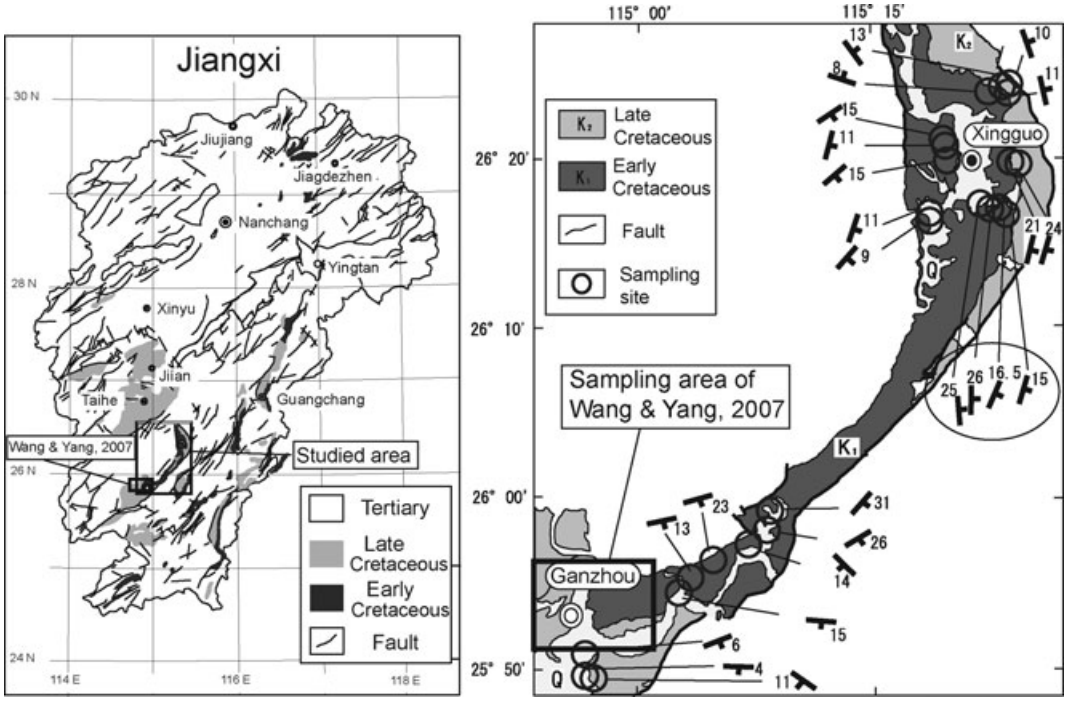


Figure 2. Simplified geological maps of (left-hand panel) Jiangxi Province and (right-hand panel) Ganzhou and Xingguo basins, showing the distribution of Cretaceous red beds and major faults. Small open circles represent the sampling locations. The attitudes of the strata (strike and dip) are also shown.

© 2009 The Authors, *GJI*, **178,** 1327–1336
Journal compilation © 2009 RAS

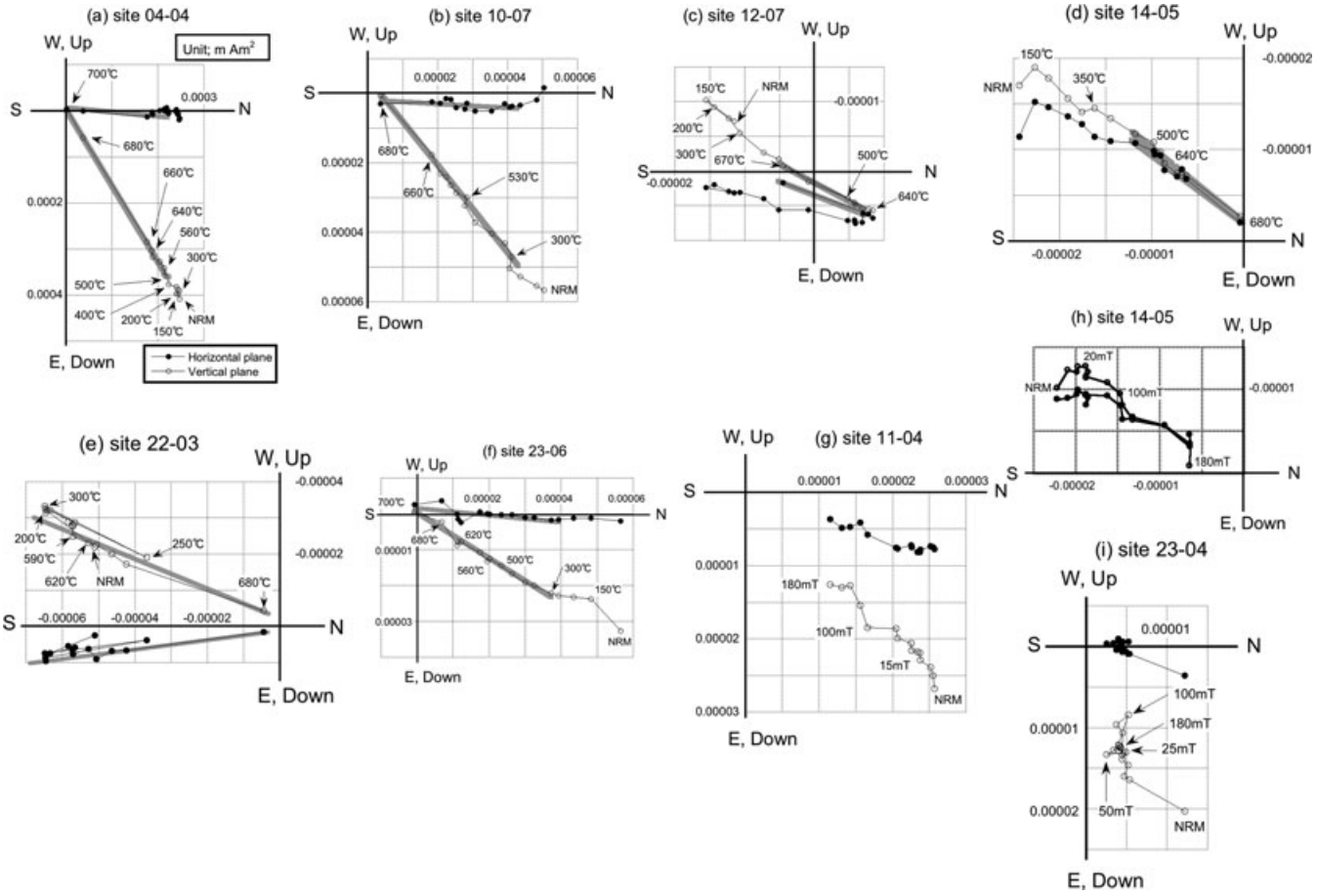


Figure 3. Representative plots of progressive thermal and alternating field demagnetization result from the early Cretaceous red sandstone samples collected in the Ganzhou and Xingguo basins. The plots are orthogonal vector projections onto the horizontal plane (solid circles) and the N–S vertical plane (open circles). *In situ* (geographic) coordinates are used.

thermal treatment (Figs 3a–f). The viscous component was eliminated at temperature steps below 300 °C for all specimens. Most samples show a single high temperature magnetization component (HTC) that is stable up to 670–700 °C. The specimens from site 12, however, showed two stable magnetization phases, with different unblocking temperatures of 590 °C (lower temperature component, LTC) and 670 °C (HTC) (Fig. 3c). Directions of normal HTCs are antipodal to those of reversed LTCs. The HTCs and LTCs found here were classified as 12H and 12L, respectively. Two specimens from site 13 showed a single normal component, but the other four specimens from the same site showed a single reversed component; we named the former 13N and the latter 13R. We excluded the 13N data when calculating the site mean directions because number of specimens used was small ($n = 2$). The specimens from sites 14, 21 and 22 showed a single reversed component with an unblocking temperature of ∼680 °C (Figs 3d and e).

We regard the HTC direction with an unblocking temperature in excess of ∼670 °C as the ChRM direction of the samples, which is useful for palaeomagnetic analysis. All the palaeomagnetic results are listed in Table 1, and the site mean directions are shown in Fig. 4. The direction-correction (DC) tilt test (Enkin 2003) gives the optimal concentration of mean directions for all sites at 51.2 ± 32.4 per cent tilt correction, although improvement of concentration of mean directions is rather unmarked ($k_{max}/k_g = 1.30$). This observation means that the remanences were not acquired immediately after sedimentation; they were acquired during sedimentary layer tilting (syntilting magnetization).

3.2 Magnetic carriers of the specimens

Stepwise AF demagnetization up to 180 mT is ineffective. This treatment was unable to reduce the remanence intensities completely, implying that high coercive force minerals mainly carry the remanences (Figs 3g–i). The IRM acquisition curves also show the presence of high coercive force minerals (Fig. 5 left-hand panel). Thermal demagnetization results of the composite IRMs show that three components (hard, medium and soft) are unblocked at 680 or 700 °C, and hard and/or medium components are dominant (Fig. 5 right-hand panel). These results indicate haematite is the dominant magnetic mineral. The specimens from site 12 showing dual polarities and site 13 have relatively high medium components (Fig. 5 right hand panel).

4 DISCUSSION

4.1 The probable explanation on the existence of dual polarities within one specimen

Specimens from site 12 showed dual and antipodal magnetic components of different unblocking temperatures (590 and 670 °C),

Table 1. Paleomagnetic results for samples of the early Cretaceous (K1) red sandstones from the Xingguo and Ganzhou basins, Jiangxi Province, China.

Site	Site	Bedding			N*	*In situ*		Untilted (51.2°)		Untilted (100°)		k	α_{95}(°)	VGP (51.2 per cent		Formation
Name	Lat. (°N)	Long. (°E)	Strike (°)	Dip (°)		Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)	Dec. (°)	Inc. (°)			Lat. (°N)	Long. (°E)	
J01	26.33	115.33	50.0	15.0	6/6	358.8	41.3	5.4	44.4	10.0	52.1	81.3	7.5	85.2	206.6	Maodian
J02	26.40	115.40	−20.0	10.0	7/5	14.1	34.6	17.1	32.4	19.1	28.7	88.5	6.5	71.9	231.0	Maodian
J03	26.40	115.40	−37.0	13.0	6/5	5.5	40.6	10.4	37.6	12.0	31.2	85.4	7.3	79.1	232.5	Maodian
J04	26.40	115.40	−13.0	11.0	7/4	12.5	34.8	16.0	32.8	18.6	29.5	25.3	12.2	72.9	232.1	Maodian
J05	26.39	115.39	−72.0	8.0	6/5	19.8	46.3	22.0	42.8	19.6	38.3	65.7	8.3	70.1	205.0	Maodian
J06	26.29	115.39	17.0	15.0	6/5	4.9	30.8	9.5	31.0	14.1	32.7	44.8	10.1	77.0	250.9	Maodian
J07	26.29	115.39	24.0	16.5	6/5	6.6	36.4	12.8	36.8	19.4	39.6	67.7	8.2	76.9	229.0	Maodian
J08	26.29	115.38	1.0	26.0	6/5	357.5	43.0	9.8	42.3	20.5	39.2	39.7	10.8	81.0	215.0	Maodian
J09	26.29	115.38	−5.0	25.0	5/4	0.1	35.1	8.8	33.5	15.8	29.4	22.3	16.6	78.6	248.1	Maodian
J10	26.19	115.32	42.0	9.0	7/7	2.2	38.9	5.8	40.3	8.7	44.3	56.4	8.1	83.8	235.5	Maodian
J11	26.27	115.32	19.0	11.0	8/7	350.5	45.3	356.1	46.9	1.6	49.6	23.9	11.6	86.1	53.8	Maodian
J12H	25.83	114.95	127.0	11.0	4/4	16.2	25.0	18.1	29.2	13.8	35.2	108.6	8.6	70.3	232.6	Maodian
J12L	25.83	114.95	127.0	11.0	6/5	194.2	−34.1	196.9	−38.4	190.4	−44.1	282.7	4.0	74.0	216.6	Maodian
J13N	25.83	114.95	92.0	4.0	2/2	No significant number of data										Maodian
J13R	25.83	114.95	92.0	4.0	4/3	192.4	−46.0	194.2	−47.1	193.3	−49.9	56.6	12.3	77.1	190.7	Maodian
J14	25.84	114.95	67.0	6.0	5/5	215.4	−29.6	217.2	−29.5	218.5	−32.6	86.1	8.3	54.0	213.7	Maodian
J15	25.91	115.04	95.0	15.0	8/7	10.6	38.0	15.9	42.5	12.3	52.8	226.3	3.7	75.6	206.7	Maodian
J16	25.92	115.06	77.0	13.0	6/6	12.5	34.7	16.9	37.5	17.8	46.2	203.0	4.7	73.8	219.2	Maodian
J17	25.94	115.08	73.0	23.0	6/5	12.4	31.0	19.6	35.3	23.9	49.9	62.6	8.5	70.8	220.6	Maodian
J18	25.96	115.12	137.0	14.0	7/6	19.7	26.9	22.4	32.3	15.1	39.2	36.8	10.1	67.6	222.7	Maodian
J19	25.97	115.13	57.0	26.0	6/6	4.0	33.3	13.0	38.0	21.5	51.8	67.5	8.2	77.2	223.8	Maodian
J20	25.98	115.13	39.0	31.0	0/6	No significant component										Maodian
J21	26.32	115.40	17.0	24.0	8/7	183.6	−26.2	189.7	−26.6	196.3	−29.2	147.4	4.6	74.8	257.0	Maodian
J22	26.32	115.40	14.0	21.0	6/6	181.6	−26.4	187.0	−26.9	192.7	−29.0	39.2	10.8	76.3	265.5	Maodian
J23	26.35	115.33	57.0	15.0	6/6	15.3	32.8	20.3	34.3	24.5	41.9	64.4	8.4	69.8	223.1	Maodian
J24	26.35	115.33	15.0	11.0	5/5	2.9	49.7	9.5	49.8	16.1	50.8	69.4	9.2	80.7	176.3	Maodian
Mean	*In situ*				23 sites (without J12L)	8.9	36.3					kg = 60.9	3.9			
	Untilted (51 per cent)				23 sites (without J12L)			12.9	38.6			$k$max = 79.3	3.4	76.3 ($k = 84.6$, $\alpha_{95} = 3.3°$)	224.3	Maximum k
	untilted (100 per cent				23 sites (without J12L)					16.9	40.3	ks = 62.1	3.9			

N^* = (number of measured specimens having significant components)/(number of collected block samples).

© 2009 The Authors, *GJI*, **178**, 1327–1336
Journal compilation © 2009 RAS

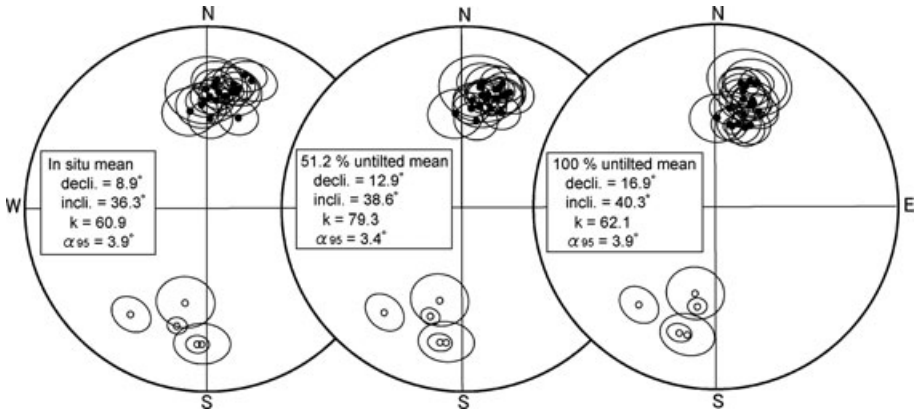


Figure 4. Equal-area projections of the site mean directions for the early Cretaceous red sandstones from the Ganzhou and Xingguo basins. The plots show *in situ* (left-hand panel), 51 per cent (central panel) and 100 per cent (right-hand panel) untilted directions. The star indicates the geocentric axial dipole field direction.

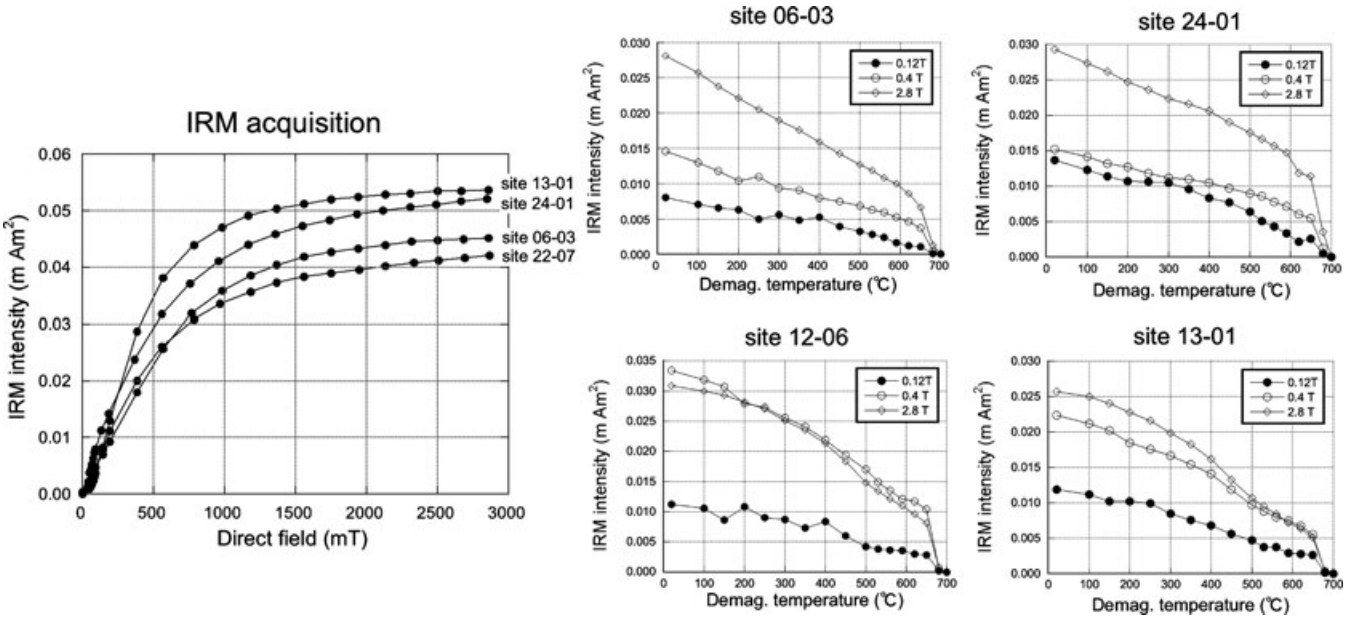


Figure 5. (left-hand panel) Isothermal remanent magnetization (IRM) acquisition curves for pilot specimens. (right-hand panels) Thermal demagnetization curves of composite IRMs (0.12, 0.4 and 2.8T in a DC field for each specimen's three perpendicular axes) for pilot specimens.

causing the direction of the normal HTC to be antipodal to that of the reversed LTC. The specimens from site 13 showed a single component of either normal or reversed polarity, and the specimens from site 14 showed a reversed polarity component. Stratigraphic levels at sites 12 and 13 were almost equal, but site 14 was situated at a lower stratigraphic level than these. These observations imply that red sandstones at these three sites recorded the Earth's magnetic field, varying from reversed to normal polarities, preserving the field during a polarity transition.

Liu & Morinaga (1999) also documented the existence of dual polarities within one specimen. This occurrence suggests that the remanence acquisition for red sandstones was completed over a much longer time than is required for polarity transition. The remanence of red sandstones, therefore, must be carried by at least two distinct magnetic phases of different unblocking temperature and/or probably of different particle sizes to record the Earth's magnetic field during polarity transition. This distinct magnetic feature shown by the specimens from sites 12 and 13 (12–07 in Fig. 3; 12–06 and 13–01 in Fig. 5, right-hand panel) seems to have played an important role in recording the Earth's magnetic field during a polarity transition. Probably, the reversed LTC was acquired in the early stage of polarity transition, and later the normal HTC was acquired.

4.2 Modelling the acquisition process of syntilting magnetization

The specimens we collected from southern Jiangxi Province showed syntilting magnetization, judged by the DC tilt test (Enkin 2003). Several previous palaeomagnetic studies from China have also reported that the optimal concentrations of site mean ChRM directions were achieved at about 50–90 per cent, indicating syntilting magnetization (for instance, Liu & Morinaga 1999; Morinaga & Liu 2004; Sun *et al*. 2006; Zhu *et al*. 2006; Wang & Yang 2007). In some of these studies, the authors have commonly believed that mean directions after incomplete untilting are approximately equal to the mean directions achieved via complete (100 per cent)

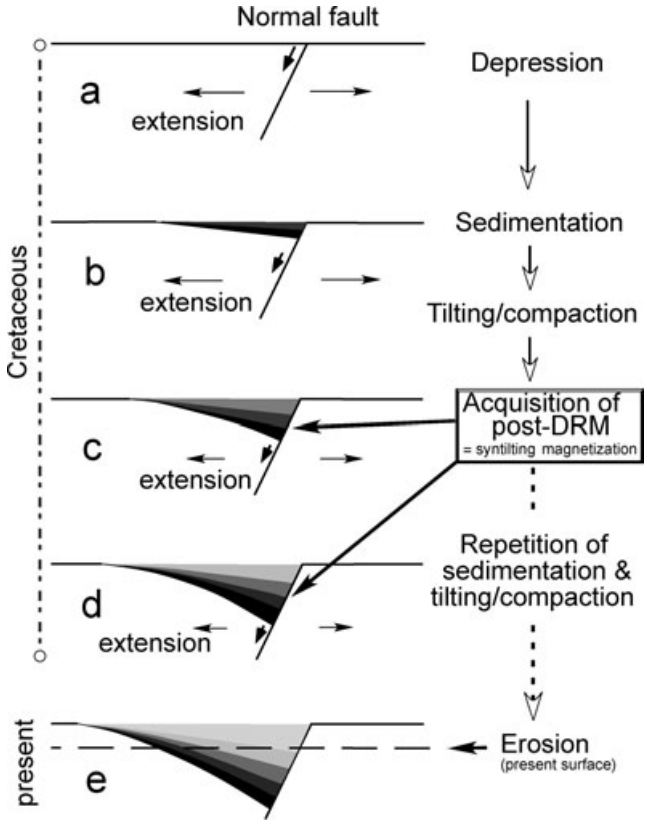


Figure 6. Schematic diagram showing the model used to explain the acquisition of syntilting magnetization. (a) Formation of a normal fault and an initial basin (depression) by extension. (b) Sedimentation, then tilting and compaction of layers by further fault movement and growth of the basin. (c) Acquisition of post-depositional remanent magnetization (DRM) and syntilting magnetization for partially tilted layers at a certain depth. (d) Repetition of sedimentation, tilting, compaction and acquisition of post-DRM during basin growth (Cretaceous period). (e) Present extensional basin surface, where we collected samples (dashed line).

untilting, and the authors have used completely untilted data for tectonic interpretation. Given the existence of many palaeomagnetic studies reporting syntilting magnetization, we should explain the acquisition process of syntilting magnetization and adopt direction data after incomplete untilting as the true direction of the palaeomagnetic field. Therefore, we suggest a model to explain the acquisition process of syntilting magnetization (Fig. 6).

A normal fault begins forming by an extension process (Fig. 6a). As a result, a sedimentary basin, called the extensional basin, forms along the normal fault, and sediments begin depositing (Fig. 6b). Sedimentary basin growth progresses contemporaneously with sedimentary layer tilting and still more sedimentation. When further sedimentation occurs on the layers deposited previously and the sedimentary bed attains a certain thickness, the weight of the upper sedimentary layers causes sedimentary compaction in the lower part of the bed. Consequently, remanence acquisition of the sedimentary bed appears to be post-depositional, and the sedimentary bed records the geomagnetic field at that time while the sedimentary layers are tilted (Fig. 6c). That is, the remanence of the sediments deposited in the extensional basin is actually syntilting magnetization acquired contemporaneously with structural and stratigraphic basin development.

In the Ganzhou and Xingguo basins, where we collected specimens for the present study, the major fault is located to the east, and the strata of almost all the sampling sites incline towards the fault. Indeed, the nearer the sampling site is to the fault, the shallower the dip of a stratum is, as partly observed in the area shown by an ellipse in Fig. 2, right-hand panel. These observations imply that these basins are extensional basins and that the model explaining the acquisition of syntilting magnetization mentioned above is applicable to these basins. Therefore, the syntilting magnetization here was acquired during the Early Cretaceous when the Ganzhou and Xingguo basins developed. The existence of specimens showing reversed polarity also supports that the syntilting magnetization was acquired before the Cretaceous Long Normal Superchron.

4.3 Implications of the early Cretaceous palaeomagnetic pole found in Jiangxi Province

The virtual geomagnetic pole (VGP) was calculated for each sampling site in the Ganzhou and Xingguo basins using the mean directions of the 51.2 per cent untilted data, because we believe that the abovementioned model can explain syntilting magnetization. The mean palaeomagnetic pole calculated from the VGPs of all sites is 76.3°N, 224.3°E ($K = 84.6$ and $\alpha_{95} = 3.3°$). This new palaeomagnetic pole for the Early Cretaceous (K1) is shown with the previously reported Cretaceous poles in Fig. 7, left-hand side. The data of these poles are also listed in Table 2. The poles from Anhui and Shandong Provinces were obtained from the areas situated in northern SCB and to the east of the Tan-Lu fault (Gilder *et al*. 1999). Although there are some other Cretaceous palaeomagnetic results from the SCB, we do not discuss the data with sufficient number of sites, suitable treatment isolating ChRMs such as progressive thermal demagnetization and principal component analysis and/or precise tilt attitude such as dykes and granites; for instance, data from coastal region compiled by Gilder *et al*. (1993) and from Hong Kong reported by Chan (1991) and Li *et al*. (2005).

The new K1 pole from the Ganzhou and Xingguo basins agrees well with the Late Cretaceous (K2) pole from the Ganzhou Basin (74.4°N, 225.1°E, $\alpha_{95} = 5.2°$) reported by Wang & Yang (2007) (Table 2 and Fig. 7, left-hand panel). This agreement indicates that there was no local tectonic deformation within southern Jiangxi during the Cretaceous.

In this new compilation (Fig. 7, left-hand panel), the streaked distribution of Cretaceous poles along a small circle centred on the sampling region, which was first documented by Enkin *et al*. (1992), is not recognized. Sixteen poles excluding two poles with high uncertainty ($\alpha_{95} > 10°$) from Jiangsu (Kent *et al*. 1986) and Sichuan Provinces (Otofuji *et al*. 1990) fairly agree with each other (Fig. 7, left-hand panel). However, the distribution of the positions of the rest 16 poles is rather large. This large pole distribution may be explained by the idea of inclination shallowing presented by Wang & Yang (2007) and Narumoto *et al.* (2006). Although this idea needs further study, we conclude that the Jiangxi region has been part of the stable body of the SCB since the Cretaceous and follow the conclusion of Morinaga & Liu (2004) and Zhu *et al.* (2006) that a large part of the SCB (excluding a 400-km-wide swath along the RRF) has behaved as a stable body since the Cretaceous.

The global mean Cretaceous palaeomagnetic pole calculated using the abovementioned 16 positions of poles from the eastern, western and central parts of the SCB (Table 2) is located at 78.8°N, 214.4°E ($\alpha_{95} = 2.6°$). This global mean pole is worth using as a reference palaeomagnetic pole of the Cretaceous for the stable SCB body when discussing relative tectonic movements between the SCB and other tectonic units and among areas in the vicinity of

© 2009 The Authors, *GJI*, **178,** 1327–1336
Journal compilation © 2009 RAS

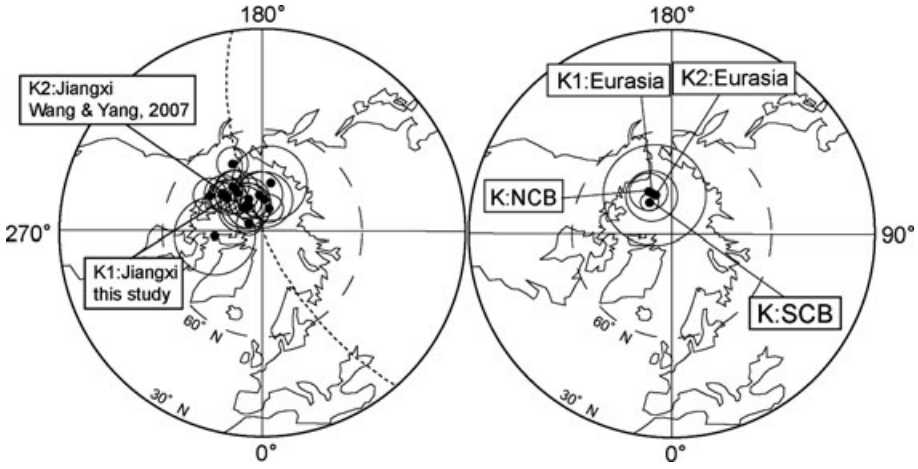


Figure 7. (left-hand panel) Cretaceous Palaeomagnetic poles from the South China Block. The colatitude line to the sampling region is also shown. (right-hand panel) Mean Cretaceous palaeomagnetic poles for the stable SCB body determined in the present study (K), stable Eurasia (K1 and K2; Enkin *et al*. 1992) and the North China Block (K; Yang & Besse 2001).

Table 2. Cretaceous paleomagnetic poles calculated from areas within the stable body of the South China Block.

Area	Location		Age	Number of sites	Pole position			Reference
(Province)	Lat. (°N)	Long. (°E)			Lat. (°N)	Long. (°E)	α95 (°)	
Eastern part								
Jiangsu*	32.0	119.0	K2	10	76.3	172.6	10.3	Kent *et al*. (1986)
Zhejiang	29.7	120.3	K1	7	76.2	225.7	4.8	Lin (1984)
Zhejiang	28.8	120.1	K1	19	81.8	207.1	6.3	Morinaga & Liu (2004)
Zhejiang	28.8	119.8	K2	19	81.0	214.3	6.4	Morinaga & Liu (2004)
Fujian	25.9	117.2	K2	22	79.7	201.7	5.8	Morinaga & Liu (2004)
Guandong	24.1	115.4	K2	9	80.8	177.7	8.0	Morinaga & Liu (2004)
Anhui	30.8	118.1	K	15	74.7	212.5	4.2	Gilder *et al*. (1999)
Shandong	37.0	120.9	K	11	81.3	217.3	5.9	Gilder *et al*. (1999)
Jiangxi	26.2	115.3	K1	23	76.3	224.3	3.3	The present study
Jiangxi	25.9	114.9	K2	14	74.4	225.1	5.2	Wang & Yang (2007)
Central part								
Hunan	28.2	110.2	K1	13	79.8	187.5	6.4	Zhu *et al*. (2006)
Hunan	28.2	110.2	K2	17	83.5	168.1	4.0	Zhu *et al*. (2006)
Hunan	26.9	112.6	K2	26	71.9	236.3	4.7	Sun et *al*. (2006)
Western part								
Sichuan	30.0	102.9	K1	23	74.5	229.0	3.7	Enkin *et al*. (1991)
Sichuan*	30.1	103.0	K	9	76.3	274.5	11.1	Otofuji *et al*. (1990)
Sichuan	26.5	102.4	K1	7	69.0	204.6	4.3	Huang & Opdyke (1992)
Sichuan	26.6	102.3	K2	18	81.5	220.9	7.1	Huang & Opdyke (1992)
Sichuan	27.9	102.3	K	33	85.2	241.7	3.5	Tamai *et al*. (2004)
Mean				16 (poles)	78.8	214.4	2.6	

*The data with $\alpha^{95} > 10^{\circ}$ were not used to calculate the mean Cretaceous pole position of the SCB.

the RRF, where local tectonic rotations and translations have been previously documented (Funahara *et al.* 1992; Gilder *et al*. 1993; Otofuji *et al*. 1998; Liu & Morinaga 1999).

The global mean pole of the stable SCB determined in the present study agrees strongly with the K1 and K2 palaeomagnetic poles for stable Eurasia (Enkin *et al*. 1992) and the Cretaceous palaeomagnetic pole for the NCB (Yang & Besse 2001; Fig. 7, right-hand panel). This agreement further supports the suggestions of Morinaga & Liu (2004) and Zhu *et al*. (2006) that no relative tectonic motion has been detected among stable Eurasia, NCB and the stable body of the SCB since the Cretaceous. In other words, the India–Asia collision did not produce relative motion among stable Eurasia, NCB and the stable body of the SCB, as already documented by Gilder & Courtillot (1997). Even if these blocks were affected by the collision, any relative motion among them cannot be detected via palaeomagnetic analysis.

5 CONCLUSIONS

From the palaeomagnetic study of Upper Cretaceous red sandstones collected at 24 sites from within the Ganzhou and Xingguo basins in southern Jiangxi Province, eastern SCB, we conclude the following main points:

(1) The specimens in the present study showed syntilting magnetization, judged by the DC tilt test (Enkin 2003). We proposed a

model to explain the acquisition process of syntilting magnetization in the extensional basin. As the model is rather likely, the syntilting magnetization obtained in the present study can be regarded as having been acquired during the Early Cretaceous when the Ganzhou and Xingguo basins developed. The existence of sites showing reversed polarity also suggests that the syntilting magnetization was acquired during the Early Cretaceous.

(2) The K1 pole from the Ganzhou and Xingguo basins agrees well with the K2 pole of the Ganzhou basin (Wang & Yang 2007), indicating that there was no local tectonic deformation within southern Jiangxi during the Cretaceous.

(3) The K1 pole from the Ganzhou and Xingguo is indistinguishable from most of the previously reported Cretaceous poles indicated by the stable body of the SCB. This observation indicates that the Jiangxi region has also been part of the stable SCB body since the Cretaceous. This conclusion supports previous findings that a large part of the SCB (excluding a 400-km-wide swath along the RRF) has behaved as a stable body since the Cretaceous (Morinaga & Liu 2004; Zhu *et al*. 2006).

ACKNOWLEDGMENTS

We are grateful to the anonymous reviewers for their very careful and constructive comments.

REFERENCES

Chan, L.S., 1991. Paleomagnetism of late Mesozoic granite intrusions in Hong Kong: implications for Upper Cretaceous reference pole of South China, *J. geophys. Res.,* **96,** 327–335.

Chen, H., Dobson, J., Heller, F. & Hao, J., 1995. Paleomagnetic evidence for clockwise rotation of the Shimao region since the Cretaceous: a consequence of India–Asia collision, *Earth planet. Sci. Lett.,* **134,** 203–217.

Enkin, R., 2003. The direction-correction tilt test: an all-purpose tilt/fold test for paleomagnetic studies, *Earth planet. Sci. Lett.,* **212,** 151–166.

Enkin, R., Courtillot, V., Xing, L., Zhang, Z., Zhuang, Z. & Zhang, J., 1991. The stationary Cretaceous paleomagnetic pole of Sichuan (South China Block), *Tectonics,* **10,** 547–559.

Enkin, R., Yang, Z., Chen, Y. & Courtillot, V., 1992. Paleomagnetic constraints on the geodynamic history of the major blocks of China from the Permian to the present, *J. geophys. Res.,* **97,** 13 953–13 989.

Funahara, S., Nishiwaki, N., Miki, M., Murata, F., Otofuji, Y & Wang, Y., 1992. Paleomagnetic study of Cretaceous rocks from the Yangtze block, central Yunnan, China, implications for the India–Asia collision, *Earth planet. Sci. Lett.,* **113,** 77–91.

Funahara, S., Nishiwaki, N., Murata, F., Otofuji, Y. & Wang, Y., 1993. Clockwise rotation of the Red River fault inferred from paleomagnetic study of Cretaceous rocks in the Shan-Thai-Malay block of western Yunnan, China, *Earth planet. Sci. Lett.,* **117,** 29–42.

Gao, Z., Chen, K. & Wei, J., 1999. *The Lithostratigraphic Dictionary of China, Multiple Classification and Correlation of the Stratigraphy of China* (in Chinese), 1st edn, China University of Geosciences Press, Wuhan, 628 pp.

Gilder, S. & Courtillot, V., 1997. Timing of the North-South China collision from new middle to late Mesozoic paleomagnetic data from the North China Block, *J. geophys. Res.,* **102,** 17 713–17 727.

Gilder, S., Coe, R., Wu, H., Kuang, G., Zhao, X., Wu, Q. & Tang, X., 1993. Cretaceous and Tertiary paleomagnetic results from Southeast China and their tectonic implications, *Earth planet. Sci. Lett.,* **117,** 637–652.

Gilder, S. *et al*., 1999. Tectonic evolution of the Tancheng-Lujiang (Tan-Lu) fault via Middle Triassic to Early Cenozoic paleomagnetic data, *J. geophys. Res.,* **104,** 15 365–15 390.

Huang, K. & Opdyke, N., 1992. Paleomagnetism of Cretaceous to lower Tertiary rocks from Southwestern Sichuan, a revisit, *Earth planet. Sci. Lett.,* **112,** 29–40.

Huang, K. & Opdyke, N., 1993. Paleomagnetic results from Cretaceous and Jurassic rocks of South and Southwest Yunnan, evidence for large clockwise rotations in the Indochina and Shan-Thai-Malay terranes, *Earth planet. Sci. Lett.,* **117,** 507–524.

Kent, D.V., Xu, G., Huang, K., Zhang, W.Y. & Opdyke, N.D., 1986. Paleomagnetism of upper Cretaceous rocks from South China, *Earth planet. Sci. Lett.,* **79,** 179–184.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Li, J.W., Zhou, M.F., Li, X.F., Fu, Z.R. & Li, Z.J., 2001. The Hunan-Jiangxi strike-slip fault system in southern termination of the Tan-Lu fault, *J. Geodyn.,* **32,** 333–354.

Li, Y., Ali, J.R., Chan, L.S. & Lee, C.M., 2005. New and revised set of Cretaceous paleomagnetic poles from Hong Kong: implications for the development of southeast China, *J. Asian Earth Sci.,* **24,** 481–493.

Lin, J.L., 1984. The apparent polar wander paths for the North and South China Blocks, pp. 248, *PhD thesis*. University of California, Santa Barbara.

Liu, Y. & Morinaga, H., 1999. Cretaceous palaeomagnetic results from Hainan Island in south China supporting the extrusion model of Southeast Asia, *Tectonophysics,* **301,** 133–144.

Lowrie, W., 1990. Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett.,* **17,** 159–162.

Mandea, M. & Macmillan, S., 2000. International geomagnetic reference field—the eighth generation, *Earth Planets Space,* **52,** 1119–1124.

Morinaga, M. & Liu, Y., 2004. Cretaceous paleomagnetism of the eastern South China Block: establishment of the stable body of SCB, *Earth planet. Sci. Lett.,* **222,** 971–988.

Narumoto, K., Yang, Z., Takemoto, K., Zaman, H., Morinaga, H. & Otofuji, Y., 2006. Anomalously shallow inclination in middle-northern part of the South China Block: palaeomagnetic study of Late Cretaceous red beds from Yichang area, *Geophys. J. Int.,* **164,** 290–300.

Otofuji, Y., Inoue, Y., Funahara, S., Murata, M. & Zheng, M., 1990. Palaeomagnetic study of eastern Tibet—deformation of the Three Rivers region, *Geophys. J. Int.,* **103,** 85–94.

Otofuji, Y., Liu, Y., Yokoyama, M., Tamai, M. & Yin, J., 1998. Tectonic deformation of the southwestern part of the Yangtze craton inferred from paleomagnetism, *Earth planet. Sci. Lett.,* **156,** 47–60.

Peltzer, G. & Tapponnier, P., 1988. Formation and evolution of strike-slip faults, rifts, and basins during the India–Asia collision: an experimental approach, *J. geophys. Res.,* **93,** 15 085–15 117.

Replumaz, A. & Tapponnier, P., 2003. Reconstruction of the deformed collision zone between India and Asia by backward motion of lithospheric blocks, *J. geophys. Res.,* **108,** doi:10.1029/2001JB000661.

Sato, K., Liu, Y., Zhu, Z., Yang, Z. & Otofuji, Y., 1999. Paleomagnetic study of middle Cretaceous rocks from Yunlong, western Yunnan, China: evidence of southward displacement of Indochina, *Earth planet. Sci. Lett.,* **165,** 1–15.

Sato, K., Liu, Y., Wang, Y., Yokoyama, M., Yoshioka, S., Yang, Z. & Otofuji, Y., 2007. Paleomagnetic study of Cretaceous rocks from Pu'er, western Yunnan, China: evidence of internal deformation of the Indochina block, *Earth planet. Sci. Lett.,* **258,** 1–15.

Schlische, R., 1992. Structural and stratigraphic development of the Newark extensional basin, eastern North America—evidence for the growth of the basin and its bounding structures, *Geol. Soc. Am. Bull.,* **104,** 1246–1263.

Sun, Z., Yang, Z., Yang, Y., Pei, J. & Yu, Q., 2006. New Late Cretaceous and Paleogene paleomagnetic results from south China and their geodynamic implications, *J. geophys. Res.,* **111**. doi:10.1029/2004JB003455.

Takemoto, K., Halim, N., Otofuji, Y., Tri, T.V., De, L.V. & Hada, S., 2005. New paleomagnetic constraints on the extrusion of Indochina: Late Cretaceous results from the Song Da terrane, northern Vietnam, *Earth planet. Sci. Lett.,* **229,** 273–285.

Tamai, M., Liu, Y., Lu, L.Z., Yokoyama, M., Halim, N., Zaman, H. & Otofuji, Y., 2004. Palaeomagnetic evidence for southward displacement

© 2009 The Authors, *GJI*, **178,** 1327–1336
Journal compilation © 2009 RAS

of the Chuan Dian fragment of the Yangtze Block, *Geophys. J. Int.,* **158,** 297–309.

Tapponnier, P., Peltzer, G., Le Dain, A.Y., Armijo, R. & Cobbold, P., 1982. Propagating extrusion tectonics in Asia: new insights from simple experiments with plasticine, *Geology,* **10,** 611–616.

The Ministry of Geology and Mineral Resources of China, 1990. *Geological Map of China* (in Chinese), Geological Publishing House, Beijing.

Wang, B. & Yang, Z., 2007. Late Cretaceous paleomagnetic results from southeastern China, and their geological implication, *Earth planet. Sci. Lett.,* **258,** 315–333.

Xu, J., Zhu, G., Tong, W., Cui, K. & Liu, Q., 1987. Formation and evolution of the Tancheng-Lujiang wrench fault system, a major shear system to the northwest of the Pacific Ocean, *Tectonophysics,* **134,** 273–310.

Yang, Z. & Besse, J., 2001. New Mesozoic apparent polar wander path for south China: tectonic consequences, *J. geophys. Res.,* **106,** 8493–8520.

Zhu, Z.W., Hao, T. & Zhao, H., 1988. Paleomagnetic study on the tectonic motion of Pan-Xi block and adjacent area during the Yinzhi-Yanshan period, *Acta Geophys. Sin.,* **31,** 420–431.

Zhu, R., Yang, Z., Wu, H., Ma, X., Huang, B., Meng, Z. & Fang, D., 1998. Paleomagnetic constraints on the tectonic history of the major blocks of China during the Phanerozoic, *Sci. China,* **41,** 1–19.

Zhu, Z., Morinaga, H., Gui, R., Xu, S. & Liu, Y., 2006. Paleomagnetic constraints on the extent of the stable body of the South China Block since the Cretaceous: new data from the Yuanma Basin, China, *Earth planet. Sci. Lett.,* **248,** 533–544.

Zijderveld, J.D.A., 1967. A.C. demagnetization of rocks: analysis of results, in *Methods in palaeomagnetism,* pp. 254–286, eds Collinson, D.W., Creer, K.M. & Runcorn, S.K., Elsevier, Amsterdam.